
University of Michigan–Ann Arbor

Department of Electrical Engineering and Computer Science

EECS 475 Introduction to Cryptography, Fall 2024

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Homework 1

Student: YOUR NAME HERE (username here)

Collaborators (if any): Name/Exercise Number(s), Name/Exercise Numbers(s)

Please refer to the course information document for important collaboration policies. Among other policies, you must list all collaborators and, moreover, cite any external resources that you have used for each exercise. You must write all solutions on your own and may not share this material outside class.

Due: September 12, 2024 (at 11:59 PM US Eastern Time)

Exercise 1

(10 points) Recall that an encryption scheme is formally defined by algorithms Gen, Enc, and Dec (respectively, the key generator, encryption, and decryption), together with a message space, cipher space, and key space. Give formal specifications for the shift cipher and the Vigenère cipher. For the latter, you may assume that the key has a length 5. Vigenère cipher is covered in the second discussion lecture. You can also find it in Section 1.1.3 from Vipul Goyal's lecture notes (see Canvas for the link).

Solution

All operations are taken mod 26

Shift Cipher Specification:

Key generator: $k \leftarrow \mathbb{Z}_{26}$

Encryption where $m = m_1 m_2 \dots m_n \in \mathbb{Z}_{26}^n$: $\text{Enc}_k(m) = (m_1 + k)(m_2 + k) \dots (m_n + k)$

Decryption where $c = c_1 c_2 \dots c_n \in \mathbb{Z}_{26}^n$: $\text{Dec}_k(c) = (c_1 - k)(c_2 - k) \dots (c_n - k) = (m_1 + k - k)(m_2 + k - k) \dots (m_n + k - k) = m_1 m_2 \dots m_n$

Vigenère Cipher Specification

Key generator: $k = k_0 k_1 k_2 k_3 k_4 \leftarrow \mathbb{Z}_{26}^5$

Encryption where $m = m_0m_1\dots m_{n-1} \in \mathbb{Z}_{26}^n$: $\text{Enc}_k(m) = c$ where $c = c_0c_1\dots c_{n-1}$ and $c_i = m_i + k_i \bmod 5$

Decryption where $c = c_0c_1\dots c_{n-1} \in \mathbb{Z}_{26}^n$: $\text{Dec}_k(c) = a_0a_1\dots a_{n-1}$ where $a_i = c_i - k_i \bmod 5 = m_i + k_i \bmod 5 - k_i \bmod 5 = m_i$

Exercise 2

(10 points) You are given three identical cards, with the exception that both sides of the first card are red, both sides of the second card are black, and one side of the third card is red and the other black. The cards are shuffled, and one card is randomly drawn and put down on the ground. If the upper side of the chosen card is red, what is the probability that the card on the ground has distinct colors on its sides?

Solution

We want to determine the probability that the other side of the card facing downward is black. Let RR , BB , and RB denote the respective events that the chosen card is the one with both its sides colored red, both its sides colored black, and that one side is red and the other is black. Let R be the event that the upturned side of the chosen card is red, as was given in the statement problem. We then have, by the law of conditional probability that

$$\Pr[RB \mid R] = \frac{\Pr[RB \wedge R]}{\Pr[R]} = \frac{\Pr[RB] \cdot \Pr[R \mid RB]}{\Pr[R]}$$

and by the law of total probability,

$$\Pr[R] = \Pr[RR] \cdot \Pr[R \mid RR] + \Pr[RB] \cdot \Pr[R \mid RB] + \Pr[BB] \cdot \Pr[R \mid BB],$$

as the probability space of our problem is comprised of RR , BB , and RB . It then follows that

$$\Pr[R] = \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot 0 = 1/3 + 1/6 = 1/2$$

and

$$\Pr[RB] \cdot \Pr[R \mid RB] = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6},$$

thus,

$$\Pr[RB \mid R] = \frac{1/6}{1/2} = \frac{1}{3}.$$

Remark: This question is basically the Monty Hall problem.

Exercise 3

(20 points)

- (a) Suppose that for some private-key encryption scheme, the ciphertext always starts with the letter “c”, can this be a perfectly secure encryption? Please explain why.
- (b) Suppose that for some private-key encryption scheme, the key always starts with the letter “c”; can this be a perfectly secure encryption? Please explain why.
- (c) Suppose a message is a string containing letters from “a” to “z”. If the ciphertext always contains the first letter of the message, can it be perfectly secure encryption? Please explain why.

Solution

- (a) Yes, this can be perfectly secure. Consider a modified one-time pad encryption scheme where Enc additionally prepends “c” to the ciphertext. Dec ignores this “c” and runs the standard decryption algorithm on the remainder.
- (b) Yes, this can be perfectly secure. Consider a modified one-time pad encryption scheme where Gen produce a key k a standard manner and prepends “c”. Enc and Dec ignore this “c”.
- (c) Yes, this can be perfectly secure. Consider a modified one-time pad encryption scheme where Enc additionally prepends “abcde...xyz” to the beginning of the ciphertext. Dec ignores the prepended value.

Exercise 4

(20 points) Let $\mathbb{Z}_n$ denote integers modulo n , and let $\mathbb{Z}_n \setminus \{0\}$ denote all the elements of $\mathbb{Z}_n$ except 0 (e.g. $\mathbb{Z}_3 = \{0, 1, 2\}$ and $\mathbb{Z}_3 \setminus \{0\} = \{1, 2\}$). For some fixed message length ℓ , let keyspace $\mathcal{K}$, message space $\mathcal{M}$, and ciphertext space $\mathcal{C}$ all be $(\mathbb{Z}_n \setminus \{0\})^\ell$ (i.e. vectors

of length ℓ over elements from $\mathbb{Z}_n \setminus \{0\}$). We partially define the following encryption scheme over $\mathbb{Z}_n \setminus \{0\}$:

$$\begin{aligned} \text{Gen}(1^\ell) &: (k_1, \dots, k_\ell) \xleftarrow{\$} (\mathbb{Z}_n \setminus \{0\})^\ell \\ \text{Enc}(k, m) &: (k_1 \cdot m_1, k_2 \cdot m_2, \dots, k_\ell \cdot m_\ell) \pmod n \end{aligned}$$

- Define a decryption algorithm for the above encryption scheme over $\mathbb{Z}_3 \setminus \{0\}$ (i.e. $\mathcal{K} = \mathcal{M} = \mathcal{C} = \mathbb{Z}_3 \setminus \{0\}$), and prove its correctness.
- Show that the encryption scheme over $\mathbb{Z}_3 \setminus \{0\}$ is perfectly secret.
- Suppose we define the encryption scheme over $\mathbb{Z}_4 \setminus \{0\} = \{1, 2, 3\}$. Is this a valid encryption scheme?

Solution

Solution here.

- We define a decryption algorithm for the Hadamard encryption scheme over $\mathbb{Z}_3 \setminus \{0\}$ for 3. For some encryption key $k = (k_1, \dots, k_\ell) \in (\mathbb{Z}_3 \setminus \{0\})^\ell$ we define

$$\text{Dec}(k, c) \mapsto (k_1^{-1} \cdot c_1, \dots, k_n^{-1} \cdot c_1).$$

Correctness holds because

$$\begin{aligned} \text{Dec}(k, \text{Enc}(k, m)) &= \text{Dec}(k, (k_1 \cdot m_1, \dots, k_n \cdot m_\ell)) \\ &= (k_1^{-1} \cdot k_1 \cdot m_1, \dots, k_n^{-1} \cdot k_n \cdot m_\ell) \\ &= (m_1, \dots, m_\ell) \\ &= m. \end{aligned}$$

Note that inverses exist for all elements in $\mathbb{Z}_n \setminus \{0\}$ for for prime n

- Suppose $\ell = 1$. For $m \in \mathcal{M}$, let

$$\text{Enc}_m(k) \mapsto \text{Enc}(k, m)$$

For arbitrary m , it is sufficient to show that Enc_m is a bijection: This implies that every ciphertext is paired with exactly one key, and is thus equally likely because every key is equally likely.

To show that Enc_m is a bijection, it is sufficient to show that it is surjective and injective. To prove surjectivity, we must show that for any $c \in \mathcal{C}$ there exists $k \in \mathcal{K}$

such that $\text{Enc}_m(k) = c$. Indeed, because all elements in $\mathbb{Z}_3$ have an inverse, we have that there exists m^{-1} such that $m \cdot m^{-1} = 1$. Thus,

$$\text{Enc}_m(m^{-1} \cdot c) = m \cdot m^{-1} \cdot c = c.$$

To prove injectivity, we must show that for $k, k' \in \mathcal{K}$, if $\text{Enc}_m(k) = \text{Enc}_m(k')$, then we must have $k = k'$. Indeed, if $\text{Enc}_m(k) = \text{Enc}_m(k')$, we have that $m \cdot k = m \cdot k'$ by definition. Multiplying both sides by m^{-1} we have that

$$m^{-1} \cdot m \cdot k = m^{-1} \cdot m \cdot k' \implies 1 \cdot k = 1 \cdot k' \implies k = k'.$$

The general case for when $\ell \geq 1$ follows because each index of the key is chosen independently at random.

- (c) The Hadamard encryption scheme over $\mathbb{Z}_4 \setminus \{0\}$ is not a valid encryption scheme. It is sufficient to show that decryption is not deterministic for $\ell = 1$. Let $k = 2$, and let $m_1 = 1, m_2 = 3$. Then we have

$$\text{Enc}(2, 1) = \text{Enc}(2, 3) = 2.$$

Exercise 5

(20 points) Prove or give a counter example: An encryption scheme with a finite message space $\mathcal{M}$ is perfectly secret if and only if for every probability distribution $\mathcal{D}$ over $\mathcal{M}$ and every $c_0, c_1 \in \mathcal{C}$ over the cipher space $\mathcal{C}$, we have $\Pr[C = c_0] = \Pr[C = c_1]$, where C is the ciphertext corresponding to a message chosen according to $\mathcal{D}$.

For this exercise, define the cipher space as the set of ciphertexts that can be returned with nonzero probability for a uniformly random message (that is, the set of ciphertexts that can ever be returned by the encryption function).

Solution

The forward direction is not true. Let $\mathcal{M} = \mathcal{K} = \{0, 1\}^n$ and $\mathcal{C} = \{0, 1\}^{n+1}$, and consider the following modification to the one-time pad:

Define the distribution $\mathbb{D}$ over $\{0, 1\}$ as $\Pr_{x \leftarrow \mathbb{D}}[x = 0] = 0.1$ and $\Pr_{x \leftarrow \mathbb{D}}[x = 1] = 0.9$.

Gen : choose $k \leftarrow \mathcal{K}$ uniformly at random

Enc(k, m) : output $c = (m \oplus k) \| b$ where $b \leftarrow \mathbb{D}$

Dec(k, c) : output $c' \oplus k$, where $c' = c[1 : n]$

In words: Enc appends a bit b from a non-uniform distribution to the encryption of

the message under the standard one-time pad and Dec decrypts the leading n bits of the ciphertext.

This scheme is indeed perfectly secret. Intuitively, appending a b from an unrelated distribution onto a ciphertext obviously cannot reveal anything more about the underlying message (relative to the original ciphertext).

Formally, for any $\tilde{c} \in C$ where $\tilde{c}[n+1] = 0$, for all $m \in \mathcal{M}$ we have

$$\begin{aligned} \Pr_{k \leftarrow \text{Gen}} [\text{Enc}(k, m) = \tilde{c}] &= \Pr_{k \leftarrow \text{Gen}} [(m \oplus k) = \tilde{c}[1:n] \wedge (b = 0)] \\ &= \Pr_{k \leftarrow \text{Gen}} [k = (m \oplus \tilde{c}[1:n])] \Pr[b = 0] \\ &= \prod_{i=1}^n \Pr[k_i = m_i \oplus \tilde{c}_i] \cdot \frac{1}{10} \\ &= \frac{1}{2^n} \cdot \frac{1}{10}, \end{aligned}$$

and for any $\tilde{c} \in C$ where $\tilde{c}[n+1] = 1$, for all $m \in \mathcal{M}$ we have

$$\begin{aligned} \Pr_{k \leftarrow \text{Gen}} [\text{Enc}(k, m) = \tilde{c}] &= \Pr_{k \leftarrow \text{Gen}} [(m \oplus k) = \tilde{c}[1:n] \wedge (b = 1)] \\ &= \Pr_{k \leftarrow \text{Gen}} [k = (m \oplus \tilde{c}[1:n])] \Pr[b = 1] \\ &= \prod_{i=1}^n \Pr[k_i = m_i \oplus \tilde{c}_i] \cdot \frac{9}{10} \\ &= \frac{1}{2^n} \cdot \frac{9}{10}, \end{aligned}$$

Thus for any given ciphertext, each message is equally likely to encrypt to it. Nevertheless, the ciphertext distribution is obviously not uniform over C . Let $c_0, c_1 \in C$ such that $c_0[n+1] = 0$ and $c_1[n+1] = 1$, and let $m \in \mathcal{M}$ be arbitrary. Then:

$$\Pr_{k \leftarrow \text{Gen}} [\text{Enc}(k, m) = c_0] = \frac{1}{2^n} \cdot \frac{1}{10} \neq \frac{1}{2^n} \cdot \frac{9}{10} = \Pr_{k \leftarrow \text{Gen}} [\text{Enc}(k, m) = c_1].$$

Perhaps another counterexample could be this: Take the usual one-time pad construction over n bits. Construct a new encryption scheme that takes n -bit plaintexts, and a $(n+2)$ -bit uniform random key. The encryption function applies the one-time pad on the plaintext using the first n bits of the key. Then it outputs the bit-wise AND of the remaining two bits of the key, so the resulting ciphertext is $n+1$ bits where the last bit has nothing to do with the plaintext. Of course, this is perfectly secret just like one-time pad is. However, given any fixed plaintext, the last bit of the ciphertext is more likely to be 0 than 1, so the ciphertext distribution is not uniform (and all possible $(n+1)$ -bit strings can appear as encryptions of some plaintext and key combination).

Exercise 6

(20 points) Consider the probability ensembles $\{X_n\}_n$ and $\{Y_n\}_n$. Consider distinguishers $\mathcal{D}_1, \dots, \mathcal{D}_m$ where $m = \text{poly}(n)$ for some polynomial function $\text{poly}(\cdot)$, such that only one can distinguish $\{X_n\}_n$ and $\{Y_n\}_n$ with non-negligible probability, but you do not know which one. Construct a distinguisher $\mathcal{D}^*$ that can distinguish between $\{X_n\}_n$ and $\{Y_n\}_n$ with non-negligible probability. Please describe your construction of $\mathcal{D}^*$ and provide a proof.

Solution

We construct distinguisher $\mathcal{D}^*$ as follows

Adversary $\mathcal{D}^*(z)$:

1. Randomly pick $\mathcal{D}_i \in \{\mathcal{D}_1, \dots, \mathcal{D}_m\}$.
2. Return $\mathcal{D}_i(z)$.

We now argue that $\mathcal{D}^*$ succeeds with non-negligible probability. That is we want the following probability to be non-negligible

$$\left| \Pr \left[z \xleftarrow{\$} \{X_n\}_n : \mathcal{D}^*(z) = 1 \right] - \Pr \left[z \xleftarrow{\$} \{Y_n\}_n : \mathcal{D}^*(z) = 1 \right] \right|$$

By definition we have that the previous equation is equivalent to

$$\left| \sum_{i \in [m]} \Pr \left[z \xleftarrow{\$} \{X_n\}_n : \mathcal{D}_i(z) = 1 \right] \cdot \Pr[\mathcal{D}^* \text{ picks } \mathcal{D}_i] - \sum_{i \in [m]} \Pr \left[z \xleftarrow{\$} \{Y_n\}_n : \mathcal{D}_i(z) = 1 \right] \cdot \Pr[\mathcal{D}^* \text{ picks } \mathcal{D}_i] \right|$$

Because $\mathcal{D}^*$ samples $\mathcal{D}_i$ randomly the previous equation is equivalent to

$$\frac{1}{m} \left| \sum_{i \in [m]} \Pr \left[z \xleftarrow{\$} \{X_n\}_n : \mathcal{D}_i(z) = 1 \right] - \Pr \left[z \xleftarrow{\$} \{Y_n\}_n : \mathcal{D}_i(z) = 1 \right] \right| \quad (1)$$

By assumption, there exist one $\mathcal{D}_j$ such that

$$\left| \Pr \left[z \stackrel{\$}{\leftarrow} \{X_n\}_n : \mathcal{D}_j(z) = 1 \right] - \Pr \left[z \stackrel{\$}{\leftarrow} \{Y_n\}_n : \mathcal{D}_j(z) = 1 \right] \right| = \mu(n)$$

for non-negligible $\mu(n)$. Therefore, because every other distinguisher distinguishes with negligible probability, for some negligible function negl , Equation (1) is equal to

$$\left| \frac{\mu(n) - \text{negl}(n)}{m} \right|.$$

which is non-negligible. Therefore, $\mathcal{D}^*$ succeeds with a non-negligible probability.

Alternative Solution

We construct distinguisher $\mathcal{D}^*$ as follows

Adversary $\mathcal{D}^*(z)$:

1. run each $\mathcal{D}_i \in \{D_1, \dots, D_m\}$ on input z .
2. Return the result of the majority.

We now argue that $\mathcal{D}^*$ succeeds with non-negligible probability. We analyze the probability that predictor wins in the predictor definition. This happens if the negligible distinguishers have more than a majority vote or the distinguishers are just under tied or tied and the non-negligible distinguisher, let's call it $\hat{D}$, outputs 1. Assume m is odd to make the calculations easier. The even case is similar.

$$\begin{aligned} \Pr(\mathcal{D}^* \text{ wins}) &= \Pr \left[(\text{sum of } D_i \neq \hat{D} \text{ correct}) \geq \lceil \frac{m}{2} \rceil \right] \\ &\quad + \Pr \left[(\text{sum of } D_i \neq \hat{D} \text{ correct}) = \lfloor \frac{m}{2} \rfloor \text{ and } \hat{D} \text{ correct} \right] \\ &\geq \sum_{i=\frac{m+1}{2}}^{m-1} \binom{m-1}{i} \left(\frac{1}{2} \pm \text{negl}(n) \right)^{m-1} + 2 \binom{m-1}{\frac{m-1}{2}} \left(\frac{1}{2} \pm \text{negl}(n) \right)^{m-1} * \left(\frac{1}{2} + \frac{1}{\text{poly}(n)} \right) \\ &\geq \frac{1}{2} \pm \text{negl}(n) - \binom{m-1}{\frac{m-1}{2}} \left(\frac{1}{2} \right)^{m-1} + 2 \binom{m-1}{\frac{m-1}{2}} \left(\frac{1}{2} \right)^{m-1} * \left(\frac{1}{2} + \frac{1}{\text{poly}(n)} \right) \\ &\geq \frac{1}{2} \pm \text{negl}(n) + \binom{m-1}{\frac{m-1}{2}} \left(\frac{1}{2} \right)^{m-1} * \frac{1}{\text{poly}(n)} \\ &\quad (\text{negl}(n) \text{ absorbs all negligible terms}) \end{aligned}$$

If $\binom{m-1}{\frac{m-1}{2}}(\frac{1}{2})^{m-1}$ is non-negligible then we are done. Using Stirling's Approximation,
 $x! \approx \sqrt{2\pi x} \frac{x^x}{e^x}$

$$\begin{aligned}
\binom{m-1}{\frac{m-1}{2}}(\frac{1}{2})^{m-1} &\approx \frac{\sqrt{2\pi(m-1)} \frac{m-1}{e}^{m-1}}{(\sqrt{2\pi \frac{m-1}{2} \frac{m-1}{2e}^{\frac{m-1}{2}}})^2} (\frac{1}{2})^{m-1} \\
&= \frac{\sqrt{2} \frac{m-1}{e}^{m-1}}{\sqrt{\pi(m-1)} \frac{m-1}{2e}^{m-1}} (\frac{1}{2})^{m-1} \\
&= \frac{\sqrt{2} \cdot 2^{m-1}}{\sqrt{\pi(m-1)}} (\frac{1}{2})^{m-1} \\
&= \Theta(\frac{1}{\sqrt{m-1}}) = \text{non-negligible}
\end{aligned}$$